Reg.No. | | | | | | | | | | | [13CE73/12CE73]

SEVENTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, DECEMBER - 2016
PRE STRESSED CONCRETE

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Assume suitably any missing data.

1. a. Write the advantages and disadvantages of prestressed concrete over Reinforced Cement Concrete. **04**

b. A pre-tensioned beam of rectangular section, 80mm wide and 120mm deep is to be designed to support concentrated loads of 4KN each at 1/3 span points over an effective span of 3m. The permissible stresses in concrete are limited to zero and 1.4N/mm^2 in tension at transfer and working load respectively. If 3mm diameter wires initially stressed to 1400 N/mm^2 are used, find the number of wires required and the eccentricity of the prestressing force assuming 20% loss in prestress. Density of concrete = 25kN/m^3. **10**

2. a. Why mild steel is not used in prestressing? **03**

b. Describe the various losses of stress that occur after the tendons are anchored in post tensioned member. **05**

c. A post tensioned cable of a beam 12m long is initially tensioned to a stress of 1000N/mm^2 at one end. If the tendons are curved so that the slope is 1in25 at each end. Find the loss of prestress due to friction given the following data: Coefficient of friction between the duct and the cable = 0.25, friction coefficient for wave effect = 0.015/mt. **06**

3. A pre-stressed concrete beam of rectangular section is 125mm wide and 250mm deep and has a span of 6m. The beam is provided with a straight tendon at a uniform eccentricity of 40mm, the pre-stressing force being 190kN. Find the deflection at the centre. **14**

i. Under the action of pre-stress and dead load of the beam.

ii. Under the action of pre-stress, dead load and live load of 3.75kN/m and including the effect of creep and shrinkage, taking the creep coefficient as 1.75.

iii. Compare these deflection with the permissible limits.

Assume E_e=38kN/mm^2, γ_c = 24kN/m^3.

4. a. Discuss different types of flexural failure. **04**

b. A post-tensioned beam with unbonded tendons is of rectangular section 400mm wide with an effective depth of 800mm. The cross sectional area of the pre-stressing steel is 2840mm^2. The effective pre-stress in the steel after all losses is 900N/mm^2. Effective span of the beam is 16m. Take f_{ck}=40N/mm^2, estimate the ultimate moment of resistance of the section using IS-1343 code recommendation. **10**

5. a. Write a brief note on stress distribution in end block with a neat sketch showing the stress distribution. **04**

b. The end block of a post-tensioned pre-stressed member is 550mm deep. Pre-stressing force in the cable is 1000kN anchored by plate anchorages, 150mm×150mm, located with their centers at 125mm from the edges of the end block. The cable dust is 50mm diameter, the characteristic cube strength is (f_{ck}) = 45N/mm^2. The cube strength of concrete at transfer (f_{ci})=25N/mm^2. The characteristic yield stress in mild steel anchorage reinforcement is 260N/mm^2. Design suitable anchorages for end block. Assume 12mm diameter bars. **10**

6. A 6m span composite beam consists of a 120mm × 210mm precast stem and a cast in situ flange 480mm × 50mm. The stress is a post-tensioned unit which is subjected to an initial pre-stressing force of 230kN. The loss of pre-stress is 15%, The tendons are provided such that their centre of gravity is 80mm above the soffit. The composite beam has to support a live load of 4kN/m. Determine the resultant stresses in the stem and flange if the beam is **14**

i. Unsupported

ii. supported

7. a. Discuss the different methods of circumferential wire winding. **04**

b. A PSC tank of diameter 20mt has to resist an internal pressure head of 5m of water. Find the reinforcement per meter of height and the thickness of concrete required. Strength of mortar coating over the wises may be ignored. Take f_{ck}=35N/mm^2 f_c= safe stress in concrete = 0.5 f_{ck} at transfer. Stress in concrete at servicing condition shall remain compressive, fs = 1000 N/mm^2, loss of pre-stress = 18%. m=8. Assume 6mm PS wire. **10**

8. a. Write a brief note on pretensiond and post-tensioned concrete bridge desk. **07**

b. Discuss the analysis of prestressed water tank for fixed base, hinged base and sliding base. **07**